Huan Li — Curriculum Vitae

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I am an Assistant Professor and EU Marie Curie IF Fellow with the Department of Computer Science, Aalborg University (AAU). My research interests include **mobile computing**, **data-intensive systems**, and **machine learning**.

Education

Doctor's Degree — Zhejiang University, China
Computer Science and Software Engineering — *Sep 2012 - Jun 2018*
Thesis: "*Uncertain Indoor Mobility Data Mining and Analytics*"; Advisors: Prof. Gang Chen and Prof. Lidan Shou.

Bachelor's Degree — Sichuan University, China
Computer Science — *Sep 2008 - Jun 2012*

Employment

Assistant Professor — Aalborg University
Department of Computer Science — *Jul 2020 - now*

Postdoctoral Research Fellow — Aalborg University
Department of Computer Science — *May 2019 - Jun 2019*

Senior Algorithm Engineer — Ant Financial, Alibaba Group
LBS Data Intelligence Team — *Jul 2018 - May 2019*

Research Assistant — Aalborg University
Department of Computer Science — *Sep 2014 - Feb 2015*

Selected Publications

✲ Journal Articles

- **Huan Li**, Hua Lu, Gang Chen, Ke Chen, Qinkuang Chen, Lidan Shou. Towards translating raw indoor positioning data into mobility semantics. *ACM/IMS Transactions on Data Science* (TDS), 1(4) No.26: 1-37, 2020.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding most popular indoor semantic locations using uncertain mobility data. *IEEE Trans. Knowl. Data Eng.* (TDKE), 31(11): 2108-2123, 2019.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In search of indoor dense regions: An approach using indoor positioning data. *IEEE Trans. Knowl. Data Eng.* (TDKE), 30(8): 1481 - 1495, 2018.

✲ Conference Papers

- **Huan Li**, Hua Lu, Lidan Shou, Ke Chen, Gang Chen. SAKE: Spatial question answering over knowledge graph based on embedding techniques. *IEEE ICDE*, 12 pages, 2021.
- **Huan Li**, Hua Lu, Muhammad Aamir Cheema, Lidan Shou, Gang Chen. Indoor mobility semantics annotation using coupled conditional Markov networks. *IEEE ICDE*, pp. 1441-1452, 2020.
- **Huan Li**, Feichao Shi, Hua Lu, Gang Chen, Ke Chen, Lidan Shou. TRIPS: A system for translating raw indoor positioning data into visual mobility semantics. *PVLDB*, 11(12):1918 - 1921, 2018.
- **Huan Li**, Hua Lu, Xin Chen, Gang Chen, Ke Chen, Lidan Shou. Vita: A versatile toolkit for generating indoor mobility data for real-world buildings. *PVLDB*, 9(13):1453 - 1456, 2016.

✲ Patents

- Payment-based location inference method and associated devices. CN110446167B, published on Jan 2021.
- Annotating and complementing indoor semantic trajectory in low-sampling positioning environment. CN109190656A, published on Jan 2019.
- Approximate search method for check-in users with non-real-time positioning data. CN108924778B, published on Nov 2018.
- Indoor continuous flow monitoring using probabilistic streaming mobility data. CN108924763A, published on Nov 2018.

Teaching

✲ Courses

- Database Development (AAU, bachelor level), *Fall 2020, Fall 2021.*

✲ **Project Supervision**

- Mobile Software Technologies (AAU, master level), *Spring 2020, Fall 2020, Spring 2021.*
- Machine Intelligence and Embedded Systems (AAU, bachelor level), *Fall 2019.*

✲ **PhD Supervision**

- Tiantian Liu (AAU, 2019 -), co-supervisor (main supervisor Hua Lu), Topic: Queries for indoor LBS.

Professional Services

✶ **Session chair** for VLDB 2020.

✶ **PC member** for SIGSPATIAL ISA 2018, MDM ALIAS 2019, SpatialDI 2021.

✶ **Journal referee** for TKDE, Journal of GIS, and Journal of Mobile Information Systems.

✶ **External reviewer** for VLDB, ICDE, UbiComp, KDD, WWW, CIKM, DASFAA, SIGSPATIAL, IEEE BigData, ICMR, PAKDD, etc.

Other Services

✶ **Volunteer leader** for VLDB 2014 and NDBC 2017.

✶ Campus ambassador of Hulu Inc. from 2012 to 2015.

✶ Monitor of the No.1 PhD Class of 2012 of Zhejiang University from 2012 to 2018.

✶ Monitor of the College's Innovation Class of Sichuan University from 2009 to 2012.

Selected Awards

✯ Outstanding Student of 41st Guards Algorithm Class of Ant Financial (2018).

✯ Candidate for Outstanding Doctoral Thesis of Zhejiang University (2018).

✯ Alibaba Big Data Contest (2016): Spatio-temporal distribution prediction of airport passengers (top-10/3455).

✯ Triple-A Graduate Award of Zhejiang University (2015).

✯ **Outstanding PhD Student Scholarship of Zhejiang University** (2015/2016).

✯ Outstanding Graduate Leader Award of Zhejiang University (2014).

✯ Award of Honor for Graduate of Zhejiang University (2013/2015/2016).

✯ **Creative Talents Award of Sichuan University** (2012).

✯ **China National Scholarship** (2009/2010/2011).

Grants

✯ **EU Marie Skłodowska-Curie Individual Fellowships** (2020-2022), 219,000 EUR.

✯ VLDB Travel Grant (2016), 1,500 USD.

✯ National University Student Innovation Program of China (2010-2012), 30,000 CNY.

Selected Projects

MALOT: Managing Mobility Data Quality for Location of Things — **PI**
H2020-MSCA-IF-2019 (Grant No. 882232) — *Apr 2020 - now*

Data Management Foundations for Indoor LBS — PI (Hua Lu)
Independent Research Fund Denmark (Grant No. 8022-00366B) — *May 2019 - now*

Management and Mining of Complex Data in Intelligent Indoor Environment — PI (Lidan Shou)
The National Science Foundation of China (Grant No. 61672455) — *Jan 2017 - Jun 2018*

Data Analysis Platform for Retail Industry — **co-PI**, PI (Lidan Shou)
Industrial Collaboration — *Mar 2016 - Dec 2017*

Management and Applications of Indoor-Space Trajectory Data — PI (Hua Lu)
The Joint Research Fund for Overseas Chinese, Hong Kong and Macao Scholars (Grant No. 61528207) — *Jan 2016 - Jun 2018*

Human-Computer Fusion based Crowd Cognition: Theory and Models — PI (Chun Chen)
The National Basic Research Program of China (973 Program) (Grant No. 2015CB352402) — *Sep 2014 - Jun 2018*